

INVESTMENT INTELLIGENCE PLATFORM

Strategic & Technical Blueprint

Phase 1 — African Market Foundation → Phase 2 — International Expansion

Covers: strategic rationale · core-out platform architecture · investment-domain ontology · technical system design · Phase 1
African market anchor

Prepared August 2026 · Working draft for internal / stakeholder review

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Executive Summary

This blueprint defines an investment-asset management platform whose core promise is transparency: every recommendation the platform makes — which stock, which bond, which fund, which opportunity — must be traceable to the data, rules, and logic that produced it. The platform is built on a single foundational anchor, adapted from the reference architecture supplied: a Single Security Master feeding a Single Data Plane, governed by a Single Accounting Engine, reasoned over by an Artificial Intelligence layer, and surfaced through ten functional domains covering the full investment lifecycle.

Phase 1 anchors the build in African capital markets — starting with the continent's most liquid and investable exchanges (Johannesburg, Nigerian Exchange Group, Egyptian Exchange, Nairobi Securities Exchange, Ghana Stock Exchange, Casablanca, and the regional BRVM) — before Phase 2 extends the same architecture to international markets. Because the core layers are market-agnostic by design, Phase 2 is a data and compliance expansion, not a re-architecture.

Four things anchor this document at the user's request: (1) the platform framework, (2) a technical system design, (3) a business/product strategy case, and (4) an investment-domain ontology and vocabulary — the shared language that makes the recommendation engine's reasoning auditable rather than opaque.

Part I — Strategic Foundation

1.1 Vision

To become the trusted, transparent layer between African investors and African capital markets — a platform where every recommendation shows its work, so that trust is earned through visible reasoning rather than claimed through brand authority.

1.2 The Market Opportunity

African equity markets are large enough to matter and fragmented enough that no single transparent, pan-African guidance layer yet exists. The continent's investable exchanges vary enormously in scale, liquidity, and market infrastructure — which is precisely why a unifying data and recommendation layer creates value: individual investors and advisors currently have to reconcile ten different data conventions, currencies, and disclosure regimes by hand.

Exchange	Country / Region	Approx. Market Cap	Note
JSE (Johannesburg Stock Exchange)	South Africa	~US\$1.5 trillion	Africa's dominant exchange; ~60% of continental equity value
NGX (Nigerian Exchange Group)	Nigeria	~US\$114 billion	Africa's 2nd–3rd largest by value; strong 2025–26 rally
EGX (Egyptian Exchange)	Egypt	~US\$74 billion	250+ listings; highest listing count on the continent
NSE (Nairobi Securities Exchange)	Kenya	~US\$25–35 billion	East Africa's primary market; Safaricom-dominated
Casablanca Stock Exchange	Morocco	Top-5 continental	North Africa's largest by several rankings
GSE (Ghana Stock Exchange)	Ghana	Mid-tier, recovering	Improving macro stability supporting listings

Exchange	Country / Region	Approx. Market Cap	Note
BRVM	8 UEMOA states (West Africa)	~US\$30 billion	Only true multi-country regional exchange in Africa; CFA franc (XOF)

Figures are approximate, sourced from mid-2026 market data, and will move; they should be refreshed at build time rather than treated as fixed. The strategic point they support is durable: five or six exchanges account for most investable liquidity, which is enough to design a credible Phase 1 without waiting for full-continent coverage.

1.3 The Problem the Platform Solves

- **Opacity:** most retail and semi-professional investors receive recommendations (from apps, brokers, or advisors) with no visible reasoning — a "buy" with no evidence trail.
- **Fragmentation:** African market data lives in inconsistent formats across exchanges, currencies, and regulators, making cross-market comparison hard even for professionals.
- **Data sparsity:** many African-listed companies have thin analyst coverage, irregular disclosure, and low trading liquidity — generic global recommendation models built for developed markets misfire here.
- **Regulatory complexity:** compliance obligations differ by jurisdiction (FSCA, SEC Nigeria, CMA Kenya, FRA Egypt, and others), which most platforms bolt on late rather than design in from the start.

1.4 Value Proposition — Transparency as the Product

The platform's differentiator is not a smarter black-box model; it is that every recommendation is explainable by construction. Section 4.3 and Section 5.2 describe how this is engineered — not bolted on as a disclaimer, but built into the data model itself, so an investor or regulator can always answer the question "why this, why now, based on what."

1.5 Target Users

- Retail and mass-affluent investors across African markets seeking clear, jargon-light guidance.
- Independent financial advisors and boutique asset managers needing institutional-grade tooling without institutional-grade cost.
- Institutional allocators (pension funds, insurers, DFIs) evaluating African opportunities who need auditable, standards-aligned data.
- Regulators and compliance functions, who benefit from a platform that treats explainability as a feature rather than an afterthought.

1.6 Two-Phase Roadmap

Phase 1 (Africa) proves the model where transparency matters most — in markets underserved by existing research infrastructure. Phase 2 (International) reuses the same Security Master, Data Plane, Accounting Engine, and AI layer, extending coverage rather than rebuilding it. The architecture in Part II is deliberately market-agnostic so this extension is additive.

Part II — Platform Framework (Architecture)

2.1 Design Philosophy: Core-Out Architecture

The platform is built from the inside out, not the outside in. A shared, single-source core sits beneath every user-facing capability, so that Portfolio Management, Risk Management, Client Reporting, and every other module all

read from — and write back to — the same ground truth. This is what prevents the classic asset-management failure mode of ten modules each keeping their own slightly-different copy of the truth.

2.2 The Four Foundational Layers

Single Security Master

The canonical, deduplicated reference record for every investable instrument — equities, bonds, ETFs, funds, and eventually alternatives — identified by international standard codes (ISIN, CFI) and mapped to local market identifiers (JSE share codes, NGX tickers, EGX codes, and so on). Nothing enters the platform without first being resolved to a single Security Master record.

Single Data Plane

The ingestion and normalization layer that reconciles pricing, corporate actions, fundamentals, and alternative data from every exchange and source into one consistent structure — regardless of the format, currency, or update frequency of the original feed.

Single Accounting Engine

One book of record for positions, transactions, valuations, P&L, and cash. Every module that touches a position — order execution, compliance, performance attribution, client reporting — reads from this single ledger, so numbers never diverge between screens.

Artificial Intelligence Layer

The reasoning layer sits above the three layers below it and uses them as verified ground truth. It does not generate recommendations from ungrounded inference — it scores, ranks, and explains using the Security Master, Data Plane, and Accounting Engine as its evidence base. This ordering (data integrity first, intelligence second) is what makes explainability possible at all.

2.3 The Ten Functional Domains

Each domain is a consumer of the four core layers, not a separate silo. The right-hand column reflects a recommended Phase 1 sequencing (elaborated in Part V).

Domain	What it does	Phase 1 priority
Portfolio Management	Constructs and monitors model and client portfolios against objectives.	Now
Data Management	Ingests, cleans, and governs market and reference data for African exchanges.	Now
Investment Accounting	Books positions, transactions, and valuations against the single ledger.	Now
Investment Compliance	Screens recommendations and trades against regulatory and mandate rules.	Now
Client Reporting	Produces investor-facing statements and explains recommendations in plain language.	Now
Risk Management	Measures exposure, concentration, currency, and volatility risk.	Next
Performance Attribution	Explains what drove returns — by sector, security, currency, or timing.	Next

Domain	What it does	Phase 1 priority
Regulatory Reporting	Produces filings required by FSCA, SEC Nigeria, CMA Kenya, FRA Egypt, and peers.	Next
Order / Deal Execution	Routes and confirms trades; requires per-exchange broker connectivity.	Later
Investment Operations	Full middle- and back-office automation (settlement, reconciliation).	Later

2.4 Engineering Transparency Into the Core

Transparency is not a UI feature added at the end — it is a property of the data model. Because every recommendation is required to cite a Security Master record, a Data Plane observation, and (where relevant) an Accounting Engine position, the evidence trail exists automatically as a by-product of how data flows through the platform, rather than as a report generated after the fact. Part III formalizes the vocabulary that makes this citation possible, and Part IV.3 describes the recommendation engine mechanics in detail.

Part III — Investment-Domain Ontology & Vocabulary

3.1 Why an Ontology Is the Right Starting Point

An ontology is simply an agreed, explicit definition of the things the platform reasons about — what a "Security" is, what distinguishes a "Position" from a "Transaction," what evidence a "Recommendation" must cite. Without this, transparency is impossible even with good intentions: two engineers, or two modules, can silently mean different things by the same word. Defining the vocabulary before the schema is standard practice in institutional investment-data programs, and it is what allows the AI layer's output to be machine-checkable rather than just plausible-sounding.

3.2 Core Entity Ontology

Entity	Definition	Aligns to
Issuer	The legal entity that issues a security.	LEI — ISO 17442
Instrument / Security	A tradeable financial instrument.	ISIN — ISO 6166; CFI — ISO 10962
Asset Class	Broad grouping: equity, fixed income, cash, alternatives.	GICS / ICB taxonomy
Market / Venue	The exchange or trading venue an instrument trades on.	MIC — ISO 10383
Portfolio	A collection of holdings managed toward a stated objective.	Internal ID
Position	The quantity of an instrument held in a portfolio at a point in time.	Derived from Accounting Engine
Transaction	An economic event that changes a position (buy, sell, transfer).	Derived; ISO 20022-aligned
Corporate Action	An issuer-driven event affecting a security (dividend,	ISO 20022 corporate actions

Entity	Definition	Aligns to
	split, rights issue).	schema
Benchmark	The reference index a portfolio or recommendation is compared against.	Index-provider ID
Risk Factor	A variable driving risk or return (currency, sector, rate, liquidity).	Internal taxonomy
Compliance Rule	A constraint governing which instruments or weights are eligible.	Internal rule ID, mapped to regulator source
Signal	A data-derived input used by the recommendation engine (valuation, momentum, quality).	Internal, versioned
Recommendation	The engine's output — a suggested action tied to explicit evidence.	Internal, versioned
Evidence / Rationale	The specific signals, rules, and data points cited to justify a Recommendation.	Linked to Signal + Compliance Rule

3.3 Aligning to Recognized Standards

Rather than inventing a private vocabulary, the ontology maps onto standards already used across global capital markets, which both speeds integration with African exchanges and keeps Phase 2 international expansion painless:

- ISIN (ISO 6166) — unique identifier for securities.
- CFI code (ISO 10962) — classifies an instrument's type and attributes.
- LEI (ISO 17442) — Legal Entity Identifier for issuers and counterparties.
- MIC (ISO 10383) — Market Identifier Code for exchanges and trading venues.
- GICS / ICB — sector and industry classification for asset classes and screening.
- ISO 20022 — messaging standard covering transactions and corporate actions.
- FIBO (Financial Industry Business Ontology, EDM Council) — a useful reference model for how these entities formally relate to one another.

3.4 Vocabulary Governance — Best Practice

- A single owned glossary (the table in Section 3.2 and Appendix A) is the source of truth; no module defines its own private terms for the same concept.
- Every term carries a version number; changes go through review rather than silent redefinition, since a redefined term can silently change what a past recommendation "meant."
- Where local market convention differs from the international standard (for example, a JSE-specific instrument attribute), the mapping is recorded explicitly rather than overridden.
- The glossary is exposed to the AI layer as structured context, not left implicit — this is what lets the natural-language explanation shown to an investor be generated from, and checked against, the same definitions the engine used to compute the recommendation.

Part IV — Technical System Design

4.1 Reference Architecture — Logical Layers

Layer	Responsibility
Ingestion	Connects to exchange feeds, regulator filings, and alternative data sources; normalizes formats and currencies.
Master Data (Security Master)	Deduplicates and resolves every instrument and issuer to one canonical record.
Data Plane / Storage	Time-series pricing, fundamentals, and corporate actions; the shared read layer for all modules.
Accounting Engine	Double-entry ledger for positions, transactions, valuations, and cash.
AI / Recommendation Layer	Scoring, ranking, and explanation generation, grounded in the layers above.
API Gateway	Exposes the ten functional domains to front-end apps and, later, third parties.
Presentation	Web and mobile applications for investors, advisors, and compliance teams.

4.2 Data Model Overview — the Platform's Own Knowledge Graph

At the center sits the Security Master (Issuer → Instrument), extended by Market/Venue for where it trades. Portfolios hold Positions, which are the accumulated result of Transactions and Corporate Actions booked against the Accounting Engine. The AI layer reads Signals computed from the Data Plane, checks them against Compliance Rules tied to the investor's jurisdiction, and emits a Recommendation whose Evidence explicitly references the Signals and Rules involved — giving a direct, queryable link between any recommendation and the data that produced it.

The platform implements this data model as its own knowledge graph, not only as relational tables. The Part III ontology's entities become node labels, the relationships between them become directed, named edges, and both nodes and edges carry properties — the nodes/relationships/properties (property-graph) approach detailed in Appendix C. This matters operationally: a relational join can tell you two facts are related, but a graph can be walked, step by step, along the exact chain of reasoning behind any recommendation. "Why was this recommended" becomes a traversable path through the platform's own graph rather than a narrative written after the fact — the AI layer's explanation (Section 4.3) is generated by describing that path in plain language, not by inventing one.

4.3 Recommendation Engine Design — Explainability by Construction

The engine is intentionally not a single opaque model. It is a layered pipeline that keeps each step inspectable:

- **Signal generation: valuation, quality, momentum, and liquidity signals computed transparently from Data Plane inputs, each independently viewable.**
- Rule screening: Compliance Rules and mandate constraints filter the eligible universe before scoring — ineligible instruments are never silently excluded without a logged reason.
- Composite scoring: signals are combined using a documented, adjustable weighting — not a black box — so the weighting itself is auditable and can be reviewed by an investment committee.

- Confidence and data-completeness flags: critical for African markets, where thin coverage on some listings should visibly lower confidence rather than be hidden.
- Natural-language explanation: an LLM-based layer translates the evidence trail into plain-language reasoning for the investor — generating the explanation from the computed evidence, not inventing a rationale independently.
- Audit log: every recommendation, its inputs, and any human override are retained and versioned, satisfying both investor trust and regulatory review.
- Graph traversal as evidence: because the underlying data is a knowledge graph (Section 4.2), the explanation step is a description of a real, walkable path — Recommendation → CITES → Signal → DERIVED_FROM → Instrument, Recommendation → CHECKED_AGAINST → ComplianceRule → ENFORCED_BY → Regulator — rather than free-text the model composes independently.

4.4 Technology Stack — Directional Guidance

These are starting-point directions to validate against team skills and vendor evaluation, not fixed commitments:

Component	Direction
Security Master & core data store	Relational database (e.g., PostgreSQL) with strict referential integrity for the master-data layer.
Knowledge Graph	Graph database (e.g., Neo4j, Amazon Neptune, or a graph layer over Postgres) storing the ontology in Part III as nodes, relationships, and properties — see Appendix C.
Time-series / market data	Purpose-built time-series store (e.g., TimescaleDB) or a cloud data warehouse for analytics workloads.
Ingestion / streaming	Event-streaming platform (e.g., Kafka or a managed equivalent) to normalize feeds from multiple exchanges.
Accounting Engine	Build a lightweight double-entry ledger for Phase 1 scope; evaluate specialist vendors only once Phase 1 volume justifies the cost.
AI / scoring services	Python-based scoring services plus an LLM (via API) for the explanation-generation step only — not for the core scoring logic.
APIs	REST or GraphQL gateway in front of all ten functional domains.
Front end	Web application (e.g., React) plus a mobile app for advisors and investors.
Infrastructure	Cloud-native, containerized; review data-residency requirements per African regulator before finalizing region.

4.5 Non-Functional Requirements

- Auditability: every recommendation and every trade must be reconstructable after the fact — a regulatory expectation in most of the target markets.
- Multi-currency support: ZAR, NGN, KES, EGP, GHS, MAD, XOF, and eventually international currencies, with explicit FX handling rather than silent conversion.
- Data residency: several African regulators have local-data expectations; confirm per-market before choosing cloud regions.
- Latency: most African markets do not require ultra-low-latency infrastructure in Phase 1 — prioritize correctness and auditability over speed.
- Security: standard controls for a regulated financial platform (encryption at rest/in transit, role-based access, segregation of duties).

Part V — Phase 1: African Market Anchor

This section deliberately draws together all four areas requested as the Phase 1 anchor: the data foundation, the recommendation logic, the full module set, and the regulatory layer — specific to African markets.

5.1 Data Foundation for Africa

Phase 1's Security Master and Data Plane are built exchange-first rather than country-first, prioritizing the markets identified in Section 1.2 by investable liquidity: JSE, NGX, EGX, NSE Kenya, GSE, Casablanca, and BRVM. Each exchange requires its own feed-normalization work (ticker conventions, settlement cycles, currency, disclosure format), which is exactly what the Single Data Plane is designed to absorb without leaking exchange-specific quirks into the modules above it.

5.2 Recommendation Logic Tuned to African Market Realities

Generic global recommendation models misfire on African markets for identifiable reasons, so Phase 1 logic should explicitly correct for them:

- Thin coverage: many listings have little or no analyst research — the confidence-flag mechanism in Section 4.3 is not optional here, it is central.
- Liquidity risk: low trading volume on some counters means a "good" recommendation can be practically unexecutable at size — position-sizing guidance should account for average daily volume, not just conviction.
- Currency risk: local-currency returns (naira, cedi, kwacha, etc.) can diverge sharply from USD returns; recommendations should show both.
- Concentration: several exchanges are dominated by one or two names (for example, Safaricom's outsized weight on the NSE) — sector and single-name concentration checks matter more here than in deep, diversified markets.

5.3 Module Rollout Sequencing

Building on the priority column in Section 2.3, a workable Phase 1 sequence:

Stage	Modules	Rationale
0–6 months	Data Management, Investment Accounting	Nothing else is trustworthy without the core data and ledger in place first.
6–12 months	Portfolio Management, Investment Compliance, Client Reporting	Enables the first investor-facing recommendation-and-explanation loop.
12–18 months	Risk Management, Performance Attribution, Regulatory Reporting	Adds institutional-grade oversight once the core loop is proven.
18+ months / Phase 2 trigger	Order/Deal Execution, Investment Operations	Deferred until broker connectivity and settlement volume justify the build.

5.4 Regulatory & Compliance Map

Market	Regulator	Note
South Africa	FSCA — Financial Sector Conduct Authority	Market-conduct regulator; successor to the Financial Services Board.
Nigeria	SEC Nigeria — Securities and Exchange Commission	Scope broadened by the Investment and Securities Act 2025, including digital assets.

Market	Regulator	Note
Kenya	CMA — Capital Markets Authority	Regulates and develops the Nairobi Securities Exchange.
Egypt	FRA — Financial Regulatory Authority	Supervises all non-bank financial markets, including EGX.
Ghana	SEC Ghana — Securities and Exchange Commission	Oversees the Ghana Stock Exchange; tightening rules on trading platforms.
Morocco	AMMC — Autorité Marocaine du Marché des Capitaux	Regulates the Casablanca Stock Exchange.
8 UEMOA states (West Africa)	CREPMF / AMF-UMOA	Single regional regulator covering the BRVM across all eight member countries.

Compliance should be modeled as data, not code: each Compliance Rule (Section 3.2) is tagged to the regulator that requires it, so a rule change in one jurisdiction never has to be hunted for across the codebase — and so the same structure extends cleanly to new regulators in Phase 2.

Part VI — Roadmap & Next Steps

6.1 Suggested Milestones

Horizon	Milestone
0–6 months	Security Master live for JSE + NGX; Accounting Engine operational; ontology (Part III) ratified as the governed glossary.
6–12 months	First explainable recommendations shipped to a pilot investor group; Investment Compliance enforcing FSCA and SEC Nigeria rules.
12–18 months	Risk, Performance Attribution, and Regulatory Reporting live; EGX, NSE Kenya, and GSE data onboarded.
18–24 months	Full African exchange coverage (incl. BRVM, Casablanca); Phase 2 international data-sourcing evaluation begins.

6.2 Phase 2 Triggers — International Expansion

Because the four core layers are market-agnostic, Phase 2 should be triggered by demand and data-partner readiness rather than a fixed date: once Phase 1 modules are stable and a credible international data source (equivalent in reliability to the African exchange feeds) is contracted, the same Security Master, Data Plane, Accounting Engine, and AI layer extend outward rather than being rebuilt.

6.3 Key Risks & Mitigations

Risk	Mitigation
Sparse/unreliable data on some African listings	Confidence-flag every recommendation; never present a low-data-quality signal as high-confidence.
Regulatory divergence across markets	Model compliance as tagged data (Section 5.4) rather than hard-coded

Risk	Mitigation
	logic.
Currency volatility distorting perceived returns	Always show local-currency and USD-equivalent returns side by side.
Overreach — trying to cover all ten modules at once	Hold the Section 5.3 sequencing; resist scope creep into Execution/Operations before the core loop is proven.
Explainability drift as the AI layer evolves	Version every Signal and Recommendation; changes to scoring logic must be reviewable, not silent.

Appendix A — Ontology Glossary (Reference Copy)

This is the same table as Section 3.2, reproduced here for standalone reference by engineering and compliance teams.

Term	Definition
Issuer	Legal entity that issues a security.
Instrument / Security	A tradeable financial instrument.
Asset Class	Broad grouping: equity, fixed income, cash, alternatives.
Market / Venue	Exchange or trading venue an instrument trades on.
Portfolio	A collection of holdings managed toward a stated objective.
Position	Quantity of an instrument held in a portfolio at a point in time.
Transaction	An economic event that changes a position.
Corporate Action	An issuer-driven event affecting a security.
Benchmark	Reference index a portfolio or recommendation is compared against.
Risk Factor	A variable driving risk or return.
Compliance Rule	A constraint governing eligible instruments or weights.
Signal	A data-derived input used by the recommendation engine.
Recommendation	The engine's output — a suggested action tied to evidence.
Evidence / Rationale	The specific signals and rules cited to justify a Recommendation.

Appendix B — Regulatory Reference (Reference Copy)

Market	Regulator
South Africa	FSCA — Financial Sector Conduct Authority
Nigeria	SEC Nigeria — Securities and Exchange Commission

Market	Regulator
Kenya	CMA — Capital Markets Authority
Egypt	FRA — Financial Regulatory Authority
Ghana	SEC Ghana — Securities and Exchange Commission
Morocco	AMMC — Autorité Marocaine du Marché des Capitaux
8 UEMOA states (West Africa)	CREPMF / AMF-UMOA (regulator of the BRVM)

Appendix C — Graph Ontology: Nodes, Relationships & Properties

This appendix expresses the Part III ontology in the form the platform actually implements it: a property graph, made of labeled nodes, directed relationships between them, and properties attached to both. This is the structure that backs the platform's own knowledge graph referenced in Section 4.2.

C.1 Why a Property Graph

- Multi-hop reasoning is native: "which recommendations are affected by a compliance-rule change in Nigeria" is a graph traversal, not a chain of joins across many tables.
- Evidence is structural, not reconstructed: a Recommendation's supporting Signals and checked ComplianceRules are direct edges from the Recommendation node — the evidence trail exists because of how the data is shaped, not as a report built afterward.
- Properties on relationships, not just nodes: a CITES edge can carry its own weight, a CHECKED_AGAINST edge can carry its own pass/fail result — capturing reasoning detail a plain relational foreign key cannot.
- Extending to Phase 2 is additive: new node labels (e.g., an international Market or Regulator) and edges plug into the same graph without redesigning the schema.

C.2 Node Labels & Properties

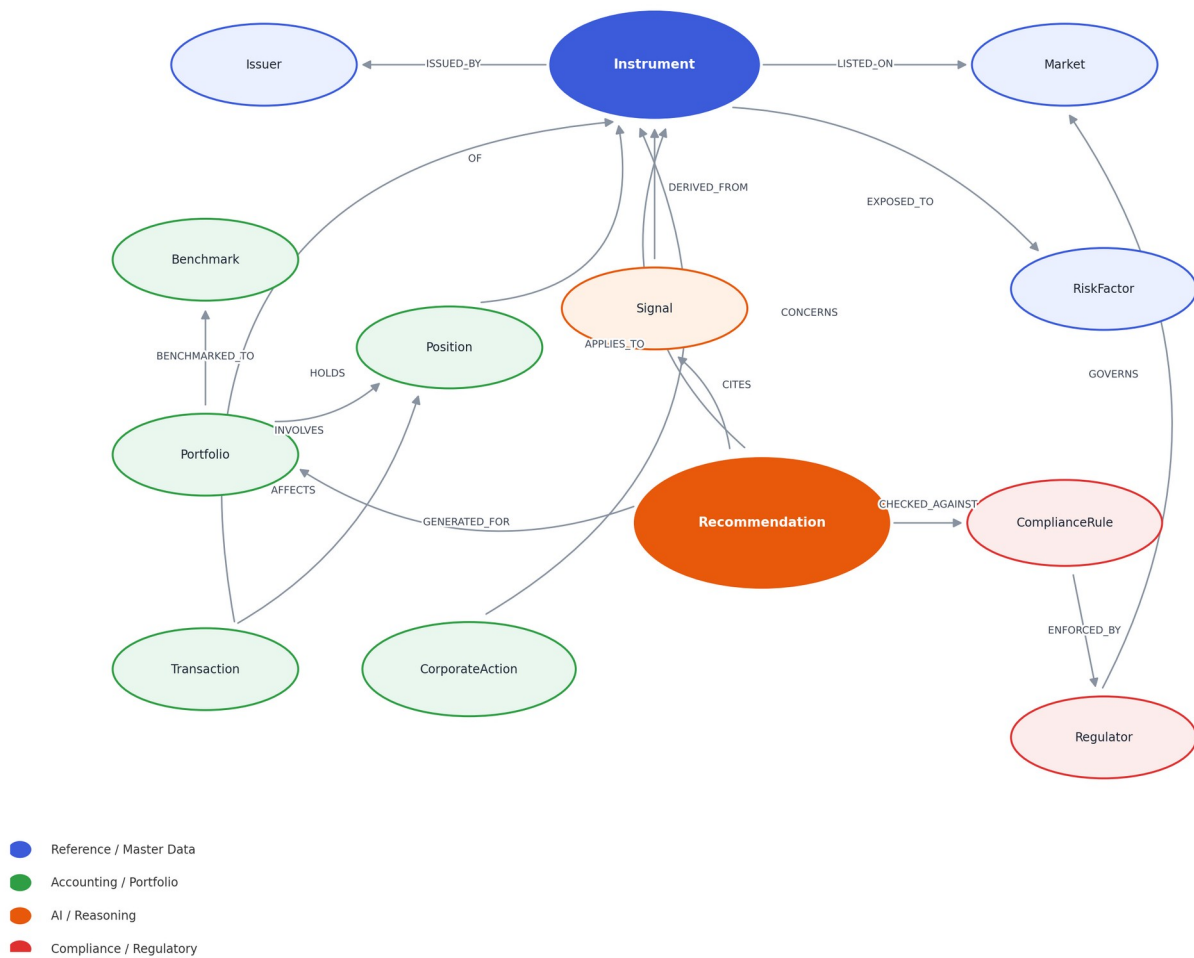
Node label	Key properties
Issuer	id, name, lei, country, sector
Instrument	isin, cfi, name, assetClass, currency
Market	mic, name, country, timezone
RiskFactor	id, name, category
Portfolio	id, name, mandate, baseCurrency
Position	quantity, asOfDate, costBasis
Transaction	id, type, quantity, price, date, currency
CorporateAction	id, type, exDate, payDate, ratio
Benchmark	id, name, provider
Signal	id, type, value, asOfDate, confidence
Recommendation	id, action, conviction, generatedAt, version

Node label	Key properties
ComplianceRule	id, description, regulator, severity
Regulator	id, name, jurisdiction

C.3 Relationships

Relationship (start → end)	Meaning	Example properties
(Instrument) -ISSUED_BY-> (Issuer)	The instrument was issued by this entity.	—
(Instrument) -LISTED_ON-> (Market)	The instrument trades on this venue.	listingDate
(Instrument) -EXPOSED_TO-> (RiskFactor)	The instrument carries this risk exposure.	weight
(Portfolio) -HOLDS-> (Position)	The portfolio currently holds this position.	—
(Position) -OF-> (Instrument)	The position is a holding of this instrument.	—
(Transaction) -AFFECTS-> (Position)	The transaction changed this position.	deltaQuantity
(Transaction) -INVOLVES-> (Instrument)	The transaction was in this instrument.	—
(CorporateAction) -APPLIES_TO-> (Instrument)	The corporate action affects this instrument.	—
(Signal) -DERIVED_FROM-> (Instrument)	The signal was computed for this instrument.	—
(Recommendation) -CITES-> (Signal)	The recommendation used this signal as evidence.	weight
(Recommendation) -CONCERNS-> (Instrument)	The recommendation is about this instrument.	—
(Recommendation) -GENERATED_FOR-> (Portfolio)	The recommendation was generated for this portfolio.	—
(Recommendation) -CHECKED_AGAINST-> (ComplianceRule)	The recommendation was screened against this rule.	result
(ComplianceRule) -ENFORCED_BY-> (Regulator)	The rule is enforced by this regulator.	—
(Regulator) -GOVERNS-> (Market)	The regulator has jurisdiction over this market.	—
(Portfolio) -BENCHMARKED_TO-> (Benchmark)	The portfolio is measured against this benchmark.	—

C.4 The Graph, Visualized



Core nodes and relationships of the platform's knowledge graph. Colors group nodes by the four foundational layers in Part II.

C.5 Example — Evidence Trail as a Graph Traversal

Answering "why was this recommended, and is it compliant" is a single traversal outward from the Recommendation node:

```
MATCH (r:Recommendation {id: $recId})
MATCH (r)-[c:CITES]->(s:Signal)-[:DERIVED_FROM]->(i:Instrument)
MATCH (r)-[:CHECKED_AGAINST]->(rule:ComplianceRule)-[:ENFORCED_BY]->(reg:Regulator)
RETURN r.action, i.name, s.type, c.weight, rule.description, reg.name
```

The result of that single traversal — which signals were cited, at what weight, and which regulator's rule was checked — is the entire evidence trail. It is the same structure the AI layer's natural-language explanation is generated from (Section 4.3), so the plain-language answer shown to an investor and the underlying graph path can always be checked against each other.